

NORMALIZATION

Normalization is the process of decomposing the larger table into smaller table by identifying their functional dependencies and removing redundancies and anomalies.

OR

The process of decomposing larger table into smaller table is known as normalization.

OR

The process of reducing a table to its normal form is known as normalization.

REDUNDANCY: It is the repetition of unwanted data.

ANOMALY: It is the side effects caused due to DML operations.

NORMAL FORM:

The table without redundancies and anomalies are said to be in normal form.

TYPES OF NORMAL FORMS:

- 1) FIRST NORMAL FORM(1NF)
- 2) SECOND NORMAL FORM(2NF)
- 3) THIRD NORMAL FORM(3NF)

NOTE: When the table is reduced to 3NF then the table is said to be normalized.

1) FIRST NORMAL FORM(1NF):

- No duplicate records.
- Multivalued data should not be present.
- EX: Table name: **QSP**

QID	NAME	COURSE
1	A	JAVA
2	B	JAVA, SQL
3	C	MT, SQL
1	A	MT



QID	NAME	C1	C2	C3
1	A	JAVA		MT
2	B	JAVA	SQL	
3	C		SQL	MT

2) SECOND NORMAL FORM:

- Table should be in **1NF**.
- Table should not have **partial functional dependency**.

EX: EMP

EMPNO	ENAME	SAL	PHNO	HIREDATE	MGR
7121	Smith	1200	9876543210	01-JAN-1982	7124
7122	Miller	1500	9876543219	15-DEC-1982	7124
7123	Scott	5500	9876543218	22-MAY-1988	7122
7124	James	3500	9876543217	23-JUL-1985	7123

3) THIRD NORMAL FORM(3NF):

- The table should be in **2NF**.
- Table should not have **transitive functional dependency**.

EX: EMP

EID	ENAME	SAL	DEPTNO	DNAME	LOC
1	A	100	10	D1	L1
2	B	200	20	D2	L2
3	C	300	10	D1	L1
4	D	400	10	D1	L1

EID >> ENAME, SAL**DEPTNO** >> DEPTNO, DNAME, LOC**R1** > (EID, ENAME, SAL)**R2** > (DEPTNO, DNAME, LOC)**R1**

EID	ENAM E	SAL
1	A	100
2	B	200
3	C	300
4	D	400

R2

DEPTN O	DNAM E	LOC
10	D1	L1
20	D2	L2